Aniruddh Murasiya

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PROFESSIONAL SUMMARY

Electrical Engineering Graduate | Electrical Building Services | LV/HV Distribution & Lighting Design | Transformer Testing & Electrical Documentation | AutoCAD, DIALux evo Basic & Bluebeam Revu Basic | MATLAB/Simulink, PSS®E & Revit | Technical Reporting & Site Support | Full Australian Working Rights

TECHNICAL SKILLS

Electrical Building Services
LV Distribution | HV/LV Systems | Power Distribution | Lighting Systems | Emergency Lighting Awareness | Switchboards | Cable Sizing | Earthing & Bonding Fundamentals | Single-Line Diagrams

Design & Documentation
Electrical Drawings | Technical Documentation | Design Reports | Equipment Schedules | Site Inspection Reports | As-Built Documentation | QA/QC Checks

Software & Tools
AutoCAD | MATLAB/Simulink | PSS®E | PSIM | PSCAD exposer | Revit | Microsoft Office | Excel | MS Project | Bluebeam Revu | DIALux evo (learning / basic)

Power Systems & Analysis
Load Flow | Fault Analysis | Renewable Integration | Transformer Manufacturing & Testing | HVDC System Modelling | Voltage Stability | Protection Fundamentals

Construction & Site Support
Site Inspections | Testing & Commissioning Support | Safety Documentation | WHS Awareness | Electrical Installation Awareness | Vendor / Workshop Coordination

Standards & Compliance
AS/NZS 3000 Awareness | Electrical Safety | Design Compliance Support | Quality Assurance | Risk Awareness

WORK EXPERIENCE

Junior Electrical Engineer | Laxmi Hi-Tech Transformers, India **Jan 2023 - May 2023**

- Supported transformer testing, final verification, and documentation activities for units up to 1000 kVA, strengthening practical understanding of electrical asset quality and compliance.
- Assisted production, tanking, and daily reporting activities for transformer units, building hands-on exposure to manufacturing workflows, inspection discipline, and schedule coordination.
- Prepared daily production and tanking reports, helping maintain traceable records and quality-focused execution.
- Built practical understanding of electrical asset behaviour, manufacturing processes, and documentation standards relevant to reliability-focused engineering roles.
- Gained practical exposure to transformer systems up to 7 MVA, supporting understanding of equipment configuration, workshop coordination, and asset-focused engineering work.

Technical Marketing Executive | Falcon Global Sales Pvt. Ltd., India **May 2022 - Nov 2022**

- Reviewed technical requirements for motors, pumps, and cables and coordinated vendors, dealers, and factory teams to identify suitable electrical solutions.
- Supported quotations, estimation, and technical documentation while helping resolve specification and supply issues across customer and manufacturing stakeholders.

Engineering Intern | Synergy Transformer, India **Jan 2022 - Apr 2022**

- Completed a 12-week hands-on internship in transformer manufacturing, testing, and assembly.
- Assisted with HV transformer assembly, terminations, wiring, and verification for units up to 15 MVA.
- Developed a strong foundation in engineering teamwork, workshop coordination, and quality-focused execution in an asset-based environment.

PROJECTS & APPLIED EXPERIENCE

Infrastructure Control System Prototype | Academic Project, Swinburne University
- Built a LabVIEW-NI DAQ control prototype integrating IR sensors, digital I/O, and real-time actuator logic.
- Developed adaptive control logic using case structures, shift registers, Boolean gating, and selector functions.
- Expanded limited NI DAQ I/O using a 74LS138 demultiplexer to maintain stable multi-output control.
- Designed a front-panel HMI for live monitoring, sensor feedback, and pedestrian override control.
- Executed integration, testing, and troubleshooting across I/O mapping, timing logic, and hardware connectivity.

Thesis - Comparative Analysis of ANN and PID Controllers for VSC-HVDC Networks | Research Project, Swinburne University
- Conducted steady-state and transient studies in MATLAB/Simulink and PSS(R)E to analyse system response under changing operating and fault conditions.
- Improved controller performance by achieving approximately 42% lower voltage overshoot and 35% faster settling time, demonstrating structured problem solving and evidence-based technical evaluation.
- Produced detailed technical reports and recommendations, strengthening capability in engineering documentation, analysis, and communicating technical findings.

Wave Energy Harvesting and Capacity Factor Optimisation | Individual Research, Swinburne University
- Evaluated renewable integration impacts, system performance, and stability considerations for grid-connected applications through simulation-based analysis.
- Strengthened understanding of decarbonisation, future energy planning, and the technical challenges of integrating renewable resources into modern power systems.

EDUCATION/QUALIFICATIONS

Master of Engineering Science, Electrical **Jul 2023 - Jul 2025**
Swinburne University of Technology, VIC
Bachelor of Engineering, Electrical **Mar 2018 - Jul 2022**
Gujarat Technological University, India

ADDITIONAL INFORMATION

Full Australian working rights | White Card holder | GradIEAust | Hackathon winner | PCB fabrication workshop | University event volunteer | Open to travel and relocation

REFERENCES

Available upon request.